

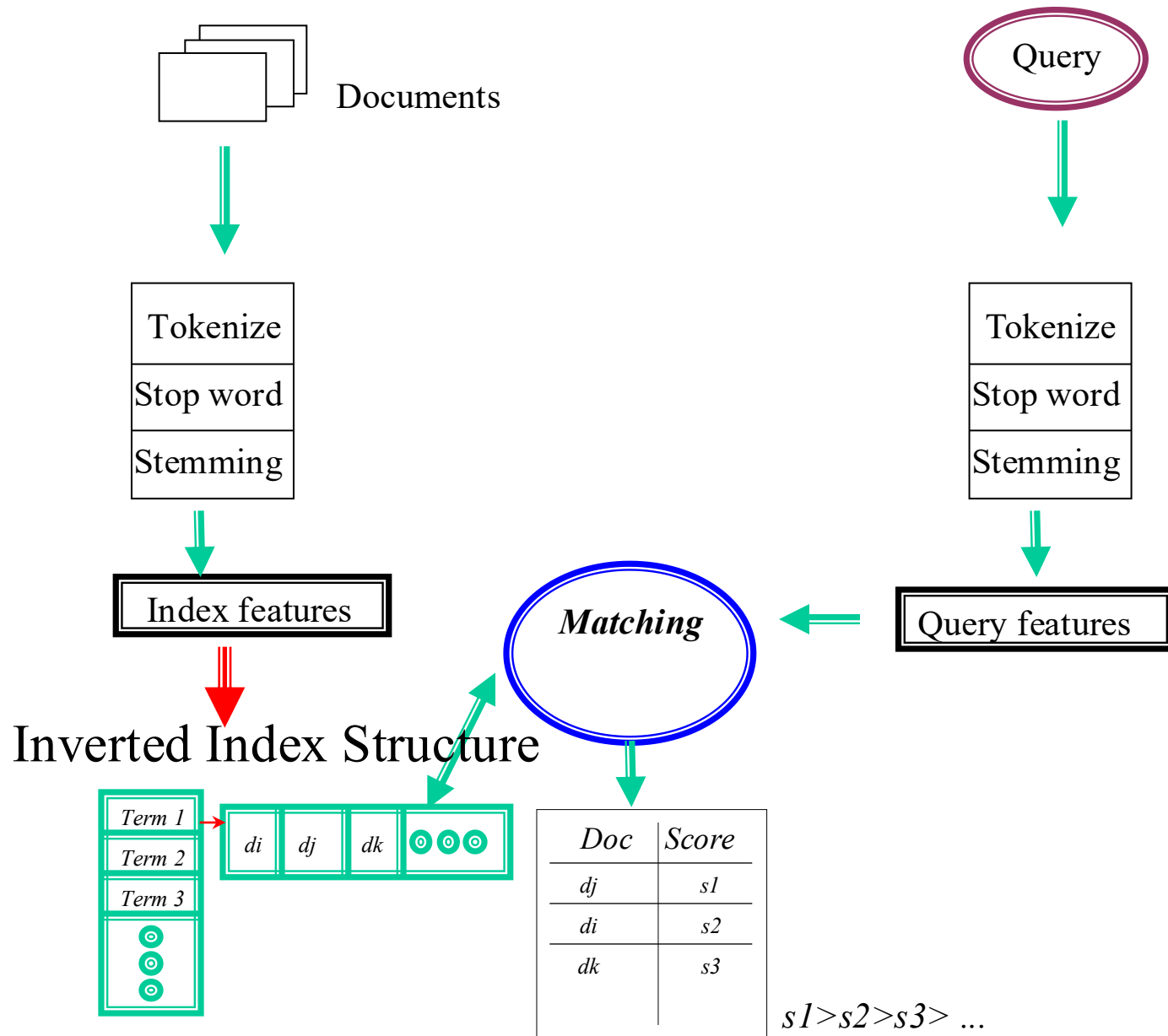
Term Weighting – Use of Statistical Information

Information Retrieval

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Explain the components of a retrieval system



Recall: Best-Match Algorithm (Co-ordinated Search)

For each document I, $\text{Score}(I) = 0$;

For each query term

Search the vocabulary list

Pull out the postings list

For each document J in the list,

$\text{Score}(J) = \text{Score}(J) + 1$

*This basically counts
how many terms are in
common between a
document and a query*

Known as **Best-Match** approach or **Co-ordinated Search**

Contrasting Assumptions

- A term is present in a document or not
 - 0 or 1 (A binary flag)
- It doesn't consider the **degree of association** between a term and a document
 - How much a document talks 'about' the 'term'?
 - The **aboutness** notion
 - This captures the semantics
 - If a term appears **often** in a document, then the document is **likely** to be **about** that 'term'
 - Indexing should capture this information

Text Statistics

- Huge variety of words used in text **but:**
- Many statistical characteristics of word occurrences are predictable
 - e.g., distribution of word counts
- **Retrieval models** and **ranking algorithms** depend heavily on statistical properties of words
 - **Document representation:** Transforming raw text of documents into a form that represents their meaning
 - Determination of a word's **semantic utility** based on its **statistical properties**
 - How can we find **meaning** in text? What does the distribution of frequency occurrences tell us about the pattern of their use?
 - e.g., **important words occur often in documents but are not high frequency in collection**

Plotting Word Frequency by Rank

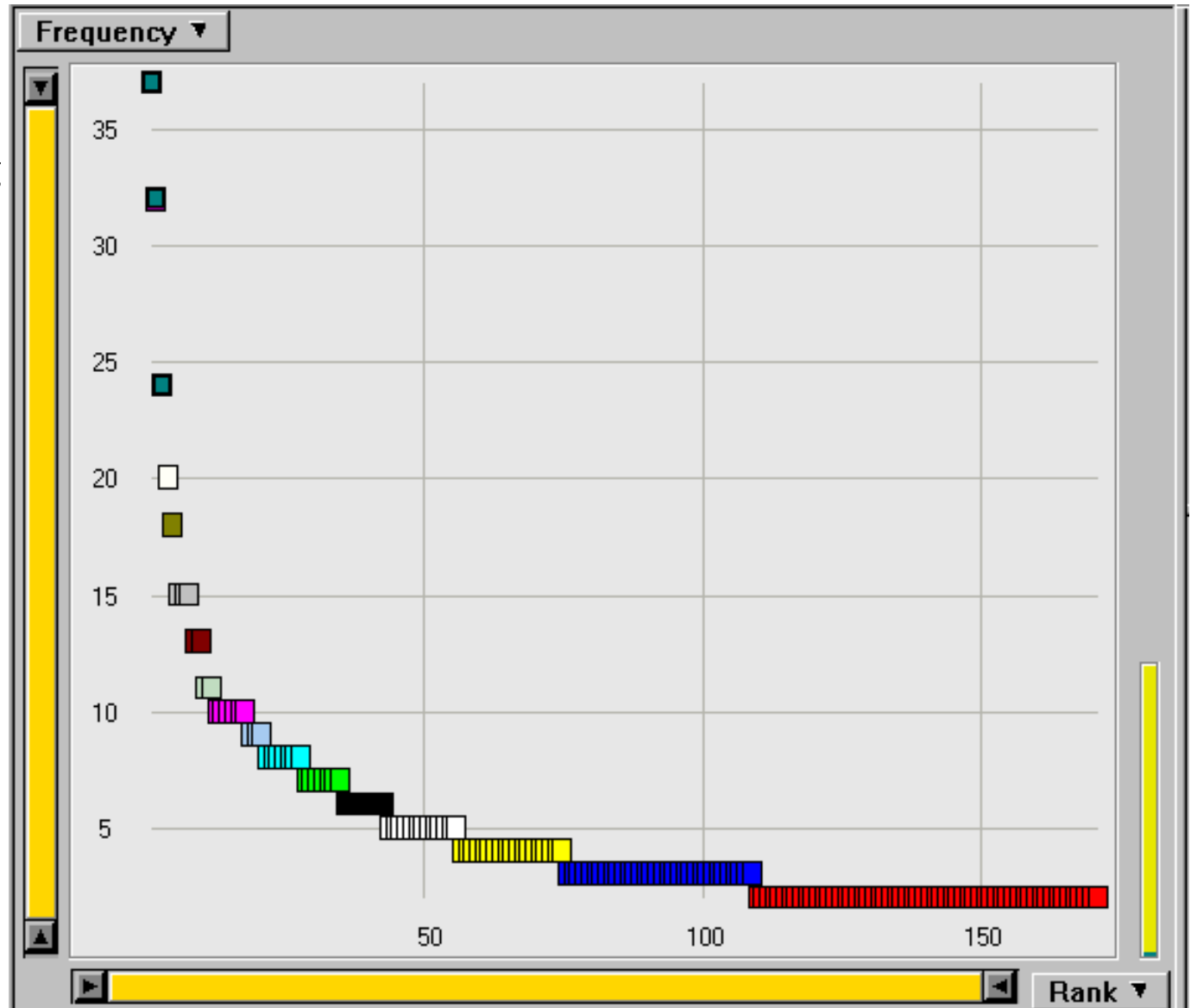
- Main idea: Count (Frequency)
 - How many times tokens occur in the text
 - Over all documents in the collection
- Now rank these according to how often they occur.

Most and Least Frequent Terms

Rank	Freq	Term			
1	37	system	150	2	enhanc
2	32	knowledg	151	2	energi
3	24	base	152	2	emphasi
4	20	problem	153	2	detect
5	18	abstract	154	2	desir
6	15	model	155	2	date
7	15	languag	156	2	critic
8	15	implem	157	2	content
9	13	reason	158	2	consider
10	13	inform	159	2	concern
11	11	expert	160	2	compon
12	11	analysi	161	2	compar
13	10	rule	162	2	commerci
14	10	program	163	2	clause
15	10	oper	164	2	aspect
16	10	evalu	165	2	area
17	10	comput	166	2	aim
18	10	case	167	2	affect
19	9	gener			
20	9	form			

The Corresponding Curve

Rank	Freq	
1	37	system
2	32	knowledg
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8	15	implem
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10	13	inform
11	11	expert
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14	10	program
15	10	oper
16	10	evalu
17	10	comput
18	10	case
19	9	gener
20	9	form

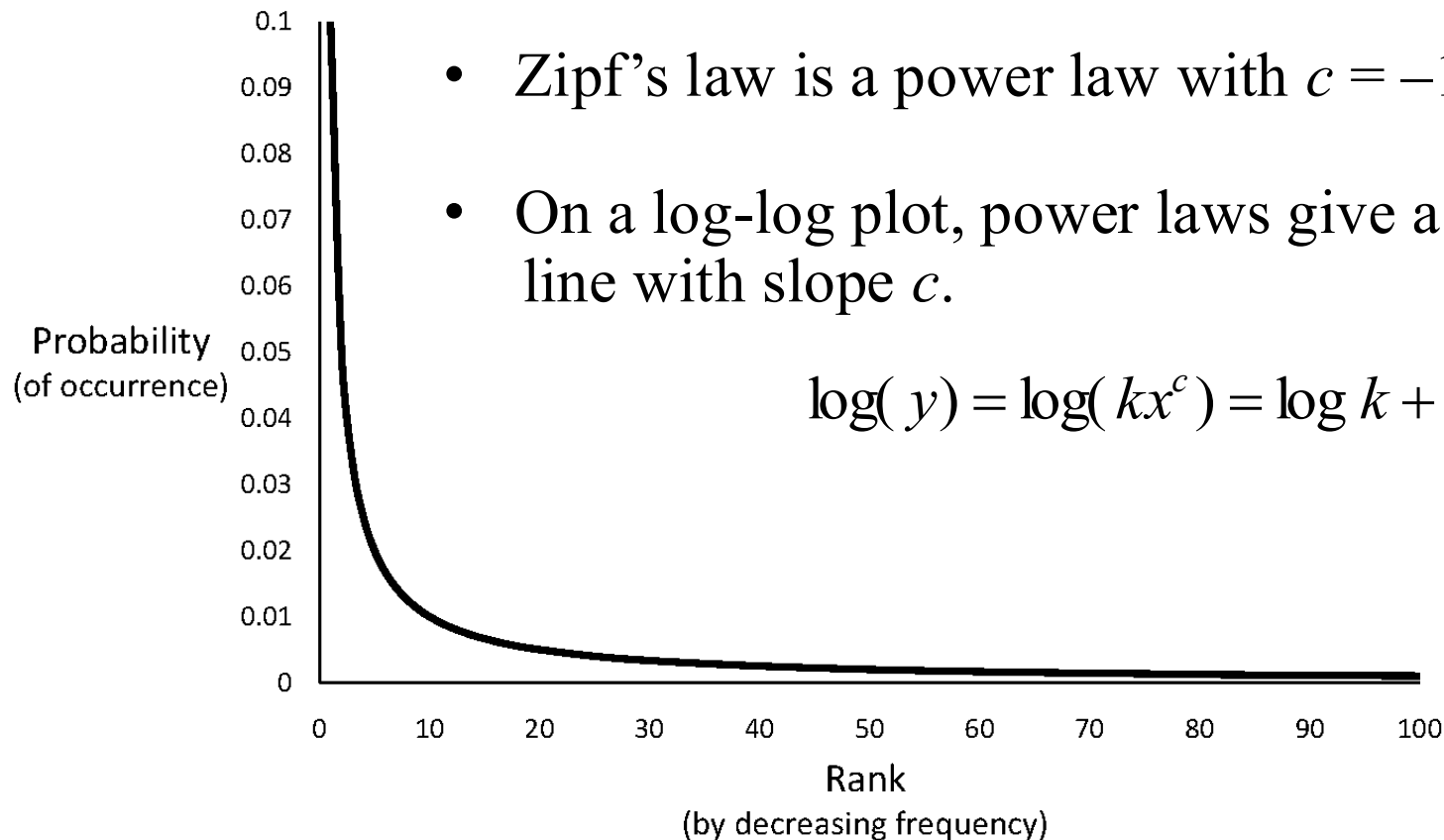


Zipf's Law

- Distribution of word frequencies is very *skewed*
 - A few words occur very often, many words hardly ever occur
 - e.g., two most common words (“the”, “of”) make up about 10% of all word occurrences in English documents
- Zipf's “law”:
 - Observation that rank (r) of a word times its frequency (f) is approximately a constant (k)
 - Assuming words are ranked in order of decreasing frequency
 - i.e., $r.f \approx k$ or $r.P_r \approx A$, where P_r is probability of word occurrence and $A \approx 0.1$ for English

Zipf's Law

- A law of the form $y = kx^c$ is called a **power law**.
- Zipf's law is a power law with $c = -1$
- On a log-log plot, power laws give a straight line with slope c .



Zipf's Distribution

- The important points:
 - A few elements occur *very frequently*
 - A medium number of elements have **medium frequency**
 - Many elements occur *very infrequently*

News Collection (AP89)

Statistics

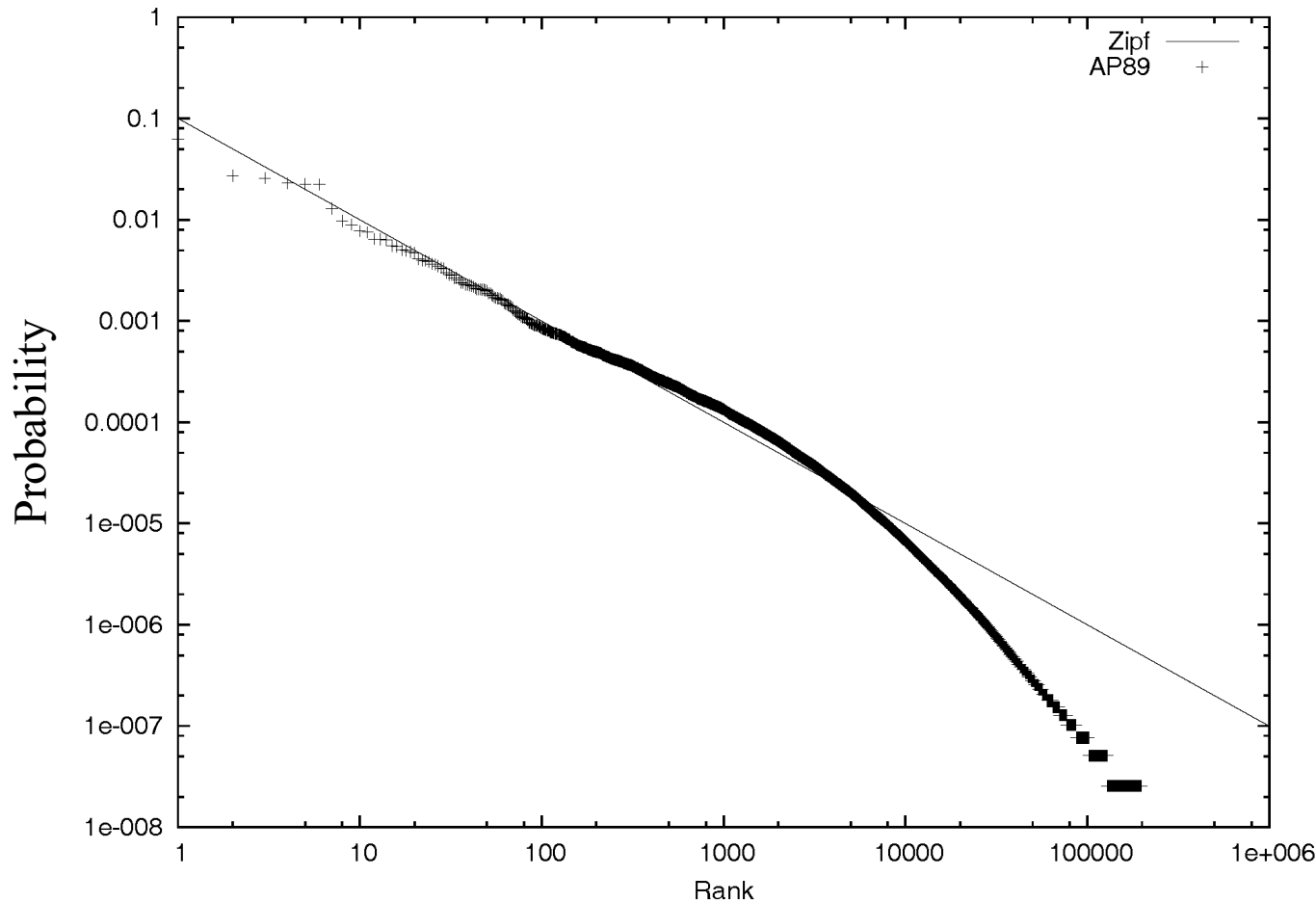
Total documents	84,678
Total word occurrences	39,749,179
Vocabulary size	198,763
Words occurring > 1000 times	4,169
Words occurring once	70,064

<i>Word</i>	<i>Freq.</i>	<i>r</i>	<i>Pr(%)</i>	<i>r.Pr</i>
assistant	5,095	1,021	.013	0.13
sewers	100	17,110	2.56×10^{-4}	0.04
toothbrush	10	51,555	2.56×10^{-5}	0.01
hazmat	1	166,945	2.56×10^{-6}	0.04

Top 50 Words from AP89

<i>Word</i>	<i>Freq.</i>	<i>r</i>	<i>P_r(%)</i>	<i>r.P_r</i>	<i>Word</i>	<i>Freq</i>	<i>r</i>	<i>P_r(%)</i>	<i>r.P_r</i>
the	2,420,778	1	6.49	0.065	has	136,007	26	0.37	0.095
of	1,045,733	2	2.80	0.056	are	130,322	27	0.35	0.094
to	968,882	3	2.60	0.078	not	127,493	28	0.34	0.096
a	892,429	4	2.39	0.096	who	116,364	29	0.31	0.090
and	865,644	5	2.32	0.120	they	111,024	30	0.30	0.089
in	847,825	6	2.27	0.140	its	111,021	31	0.30	0.092
said	504,593	7	1.35	0.095	had	103,943	32	0.28	0.089
for	363,865	8	0.98	0.078	will	102,949	33	0.28	0.091
that	347,072	9	0.93	0.084	would	99,503	34	0.27	0.091
was	293,027	10	0.79	0.079	about	92,983	35	0.25	0.087
on	291,947	11	0.78	0.086	i	92,005	36	0.25	0.089
he	250,919	12	0.67	0.081	been	88,786	37	0.24	0.088
is	245,843	13	0.65	0.086	this	87,286	38	0.23	0.089
with	223,846	14	0.60	0.084	their	84,638	39	0.23	0.089
at	210,064	15	0.56	0.085	new	83,449	40	0.22	0.090
by	209,586	16	0.56	0.090	or	81,796	41	0.22	0.090
it	195,621	17	0.52	0.089	which	80,385	42	0.22	0.091
from	189,451	18	0.51	0.091	we	80,245	43	0.22	0.093
as	181,714	19	0.49	0.093	more	76,388	44	0.21	0.090
be	157,300	20	0.42	0.084	after	75,165	45	0.20	0.091
were	153,913	21	0.41	0.087	us	72,045	46	0.19	0.089
an	152,576	22	0.41	0.090	percent	71,956	47	0.19	0.091
have	149,749	23	0.40	0.092	up	71,082	48	0.19	0.092
his	142,285	24	0.38	0.092	one	70,266	49	0.19	0.092
but	140,880	25	0.38	0.094	people	68,988	50	0.19	0.093

Zipf's Law for AP89



Zipf is quite accurate **except** for very high and low rank.

- Note problems at high and low frequencies
 - See the Mandelbrot (1954) Correction:

$$(r + \beta)^\alpha \cdot P_r = \gamma \text{ where } \beta, \alpha, \text{ and } \gamma \text{ are parameters}$$

Zipf's Law

- What is the proportion of words with a given frequency?
 - Word that occurs n times has rank $r_n = k/n$
 - Number of words with frequency n is
 - $r_n - r_{n+1} = k/n - k/(n+1) = k/n(n+1)$
 - This proportion can be found by dividing by the total number of words = highest rank = $k/1 = k$
 - So, proportion with frequency n is $1/n(n+1)$

<i>Rank</i>	<i>Word</i>	<i>Frequency</i>
1000	concern	5,100
1001	spoke	5,100
1002	summit	5,100
1003	bring	5,099
1004	star	5,099
1005	immediate	5,099
1006	chemical	5,099
1007	african	5,098

Zipf's Law

- Example word frequency ranking

<i>Rank</i>	<i>Word</i>	<i>Frequency</i>
1000	concern	5,100
1001	spoke	5,100
1002	summit	5,100
1003	bring	5,099
1004	star	5,099
1005	immediate	5,099
1006	chemical	5,099
1007	african	5,098

- To compute number of words with frequency 5,099
 - rank of “chemical” minus the rank of “summit”
 - $1006 - 1002 = 4$

Example

<i>Number of Occurrences (n)</i>	<i>Predicted Proportion ($1/n(n+1)$)</i>	<i>Actual Proportion</i>	<i>Actual Number of Words</i>
1	.500	.402	204,357
2	.167	.132	67,082
3	.083	.069	35,083
4	.050	.046	23,271
5	.033	.032	16,332
6	.024	.024	12,421
7	.018	.019	9,766
8	.014	.016	8,200
9	.011	.014	6,907
10	.009	.012	5,893

- Proportions of words occurring n times in 336,310 TREC documents
- Vocabulary size is 508,209

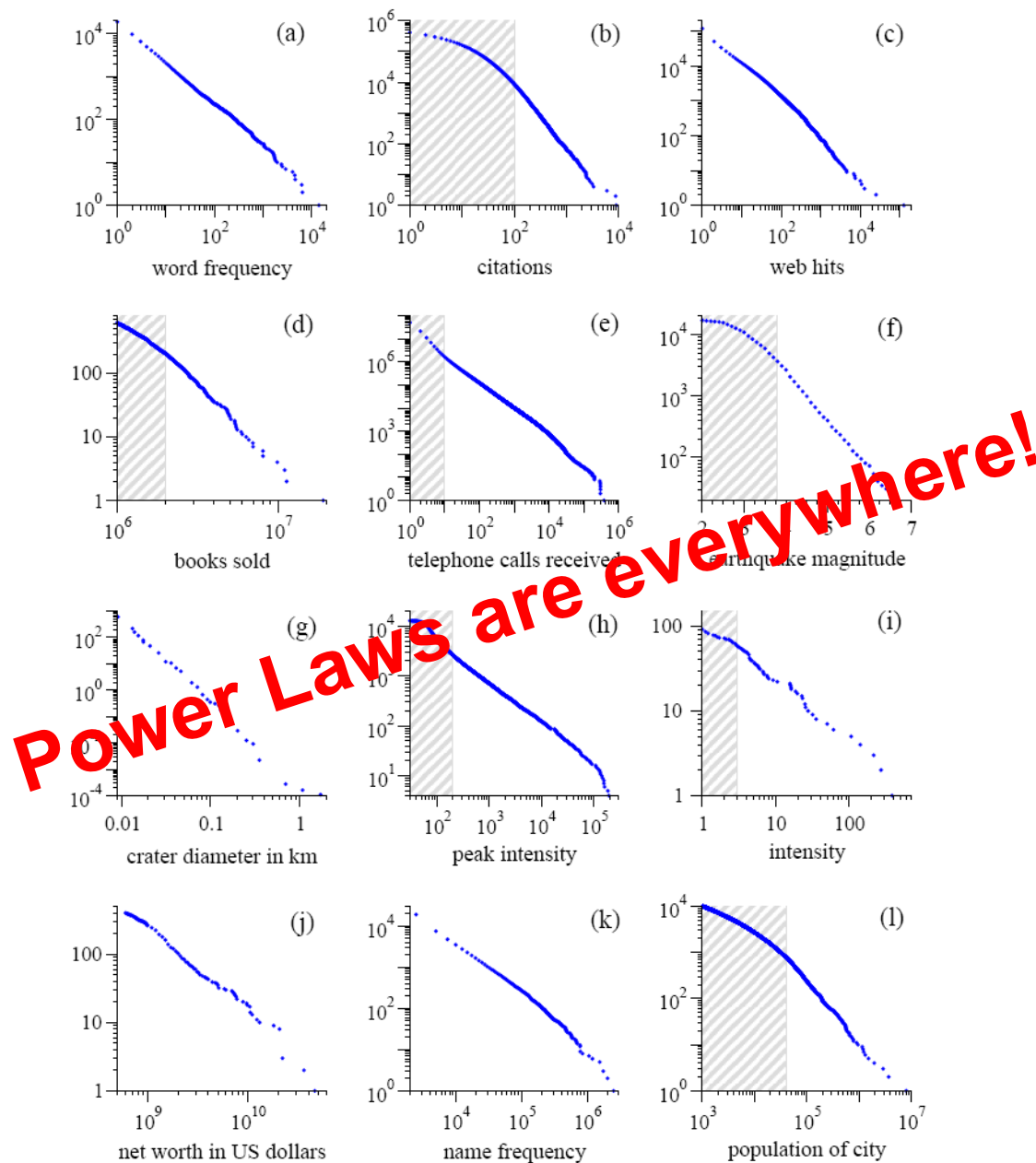


Figure from: Newman, M. E. J. (2005) "Power laws, Pareto distributions and Zipf's law." Contemporary Physics 46:323–351.

Consequences of Zipf Law

- There are always a few very frequent tokens that are not good discriminators. Referred to as “**stopwords**” in IR
 - Correspond to linguistic notion of “**closed-class**” words
 - English examples: *to, from, on, and, the, ...*
 - Grammatical classes that don’t take on new members:
you can invent a new noun (e.g., iPhone), but you can’t invent a new preposition (e.g., to).
 - Eliminating them greatly reduces inverted-index storage costs
 - Postings list for most remaining words in the inverted index will be short since they are rarer, making retrieval fast
- There are always a large number of tokens that occur **once** or very infrequently (e.g., typos)
 - These words do not describe the content of documents
- **Medium frequency** words tend to be the most **descriptive**

Vocabulary Growth

- How big is the term vocabulary?
 - How does the size of the overall vocabulary (number of unique words) grow with the size of the corpus?
- This determines how the size of the lexicon within the **inverted index** will **scale** with the size of the corpus.
- Vocabulary not really upper-bounded due to proper names, typos, invented words (e.g. product, company names), email addresses, etc.

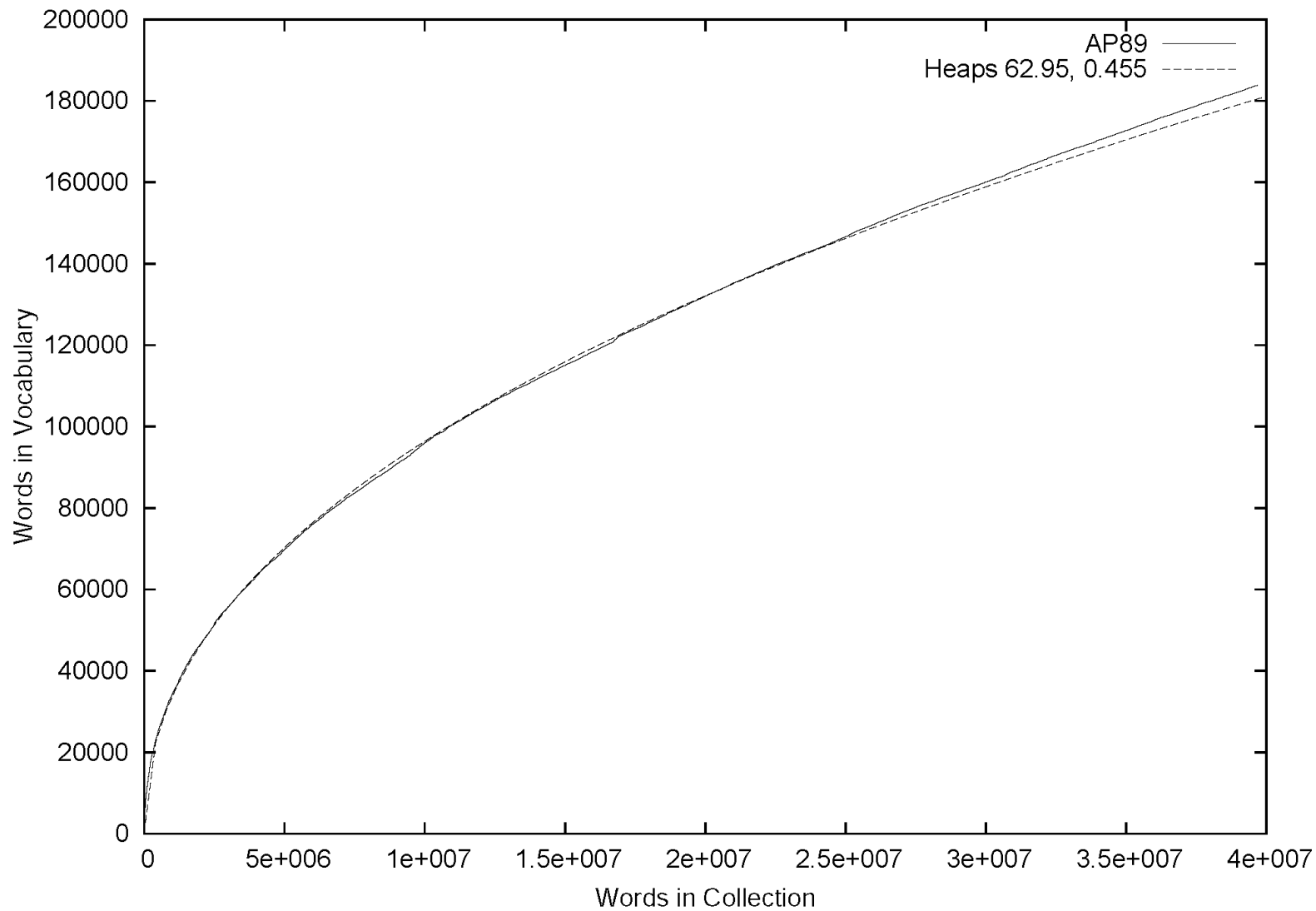
Heaps' Law

- If V is the size of the vocabulary (number of unique words) and n is the number of words in corpus:

$$V = Kn^\beta \quad \text{with constants } K, 0 < \beta < 1$$

- Typical constants (for English):
 - $K \approx 10\text{--}100$
 - $\beta \approx 0.5$

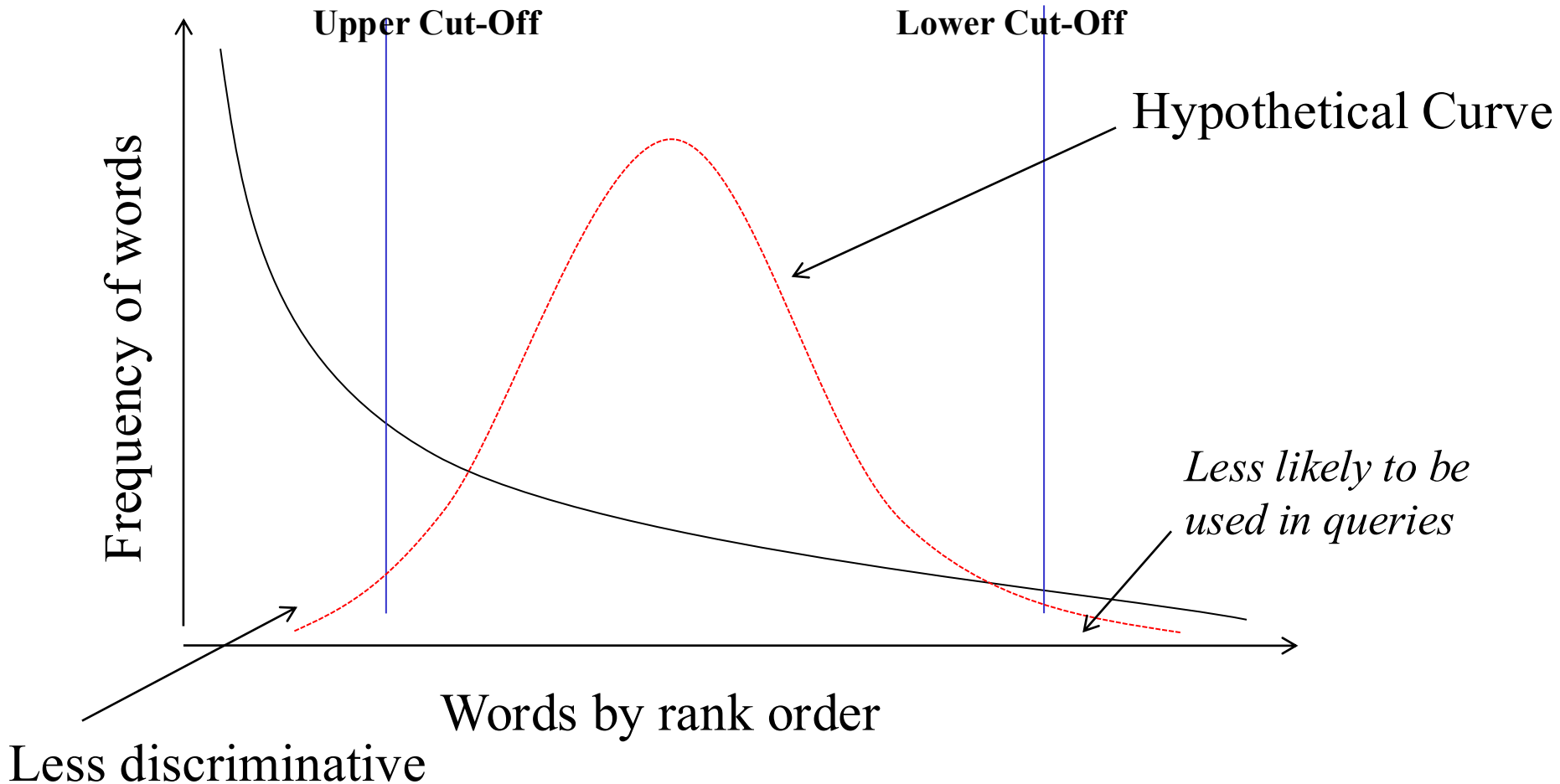
AP89 Example



Automatic Document Representation

- **Content Analysis**: transforming raw text into more computationally useful forms
 - The objective is to use a set of terms to **describe the document**
- So far we looked into properties of word occurrences
 - Word frequencies follow a Zipf distribution
 - Stopwords vs Content words
 - Word co-occurrences exhibit dependencies
 - E.g. Microsoft -> Windows; Microsoft -> Office
- Let's revisit what sorts of words are **useful** for representing document **aboutness**

Word Frequency vs. Resolving Power



The frequency of a word occurrence in an article
furnishes a useful measurement of word significance
[Luhn 1957]

Resolving Power

- Why some words occur more frequently and how such statistics can be exploited when **automatically measuring aboutness?**
 - the frequency of a word occurrence in an article furnishes a useful measurement of word significance [Luhn 1957]
- **Two critical factors**
 - Word frequency within a document
 - Collection frequency

Term Weighting

- More effective approaches score documents based on:
 - How **many** query terms they contain
 - How **discriminative** each of those terms are
- Not all terms are equally useful, so we weight them
 - Weight describes/quantifies the relationship between a keyword and a document.
 - Generality
 - Use terms with significant weights
 - Binary is a special case
 - A retrieval method can exploit these weights.

Notations

OUTPUT

- w_{kd} = weight of k^{th} word in document d

INPUTS

- f_{kd} = number of occurrences of k^{th} word in document d (term frequency)
- N = Number of documents in the collection
- D_k = number of documents containing k^{th} word

How to Weight Terms

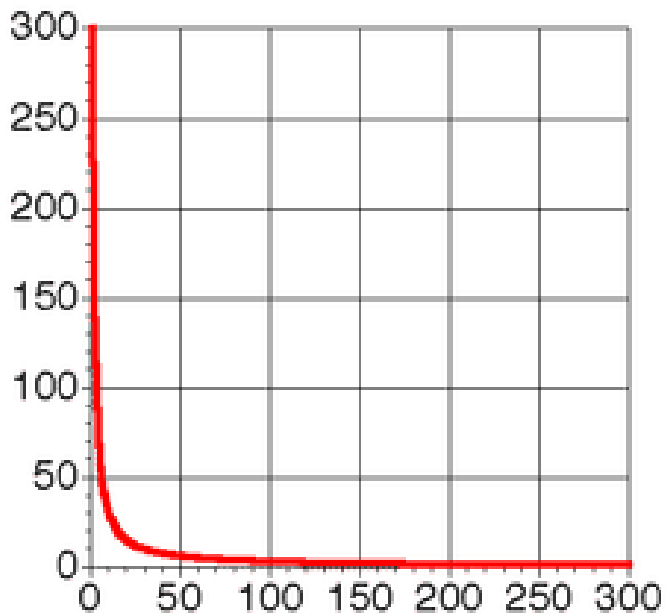
- Two demands on the weight
 - The degree to which a particular document is about a topic (or a particular keyword)
 - **Repetition** is an indication of emphasis
 - Degree to which a keyword **discriminates** documents in the collection
 - Let us call it *discrim*_k
$$w_{kd} \propto f_{kd} \times \textit{discrim}_k$$

Inverse Document Frequency (IDF)

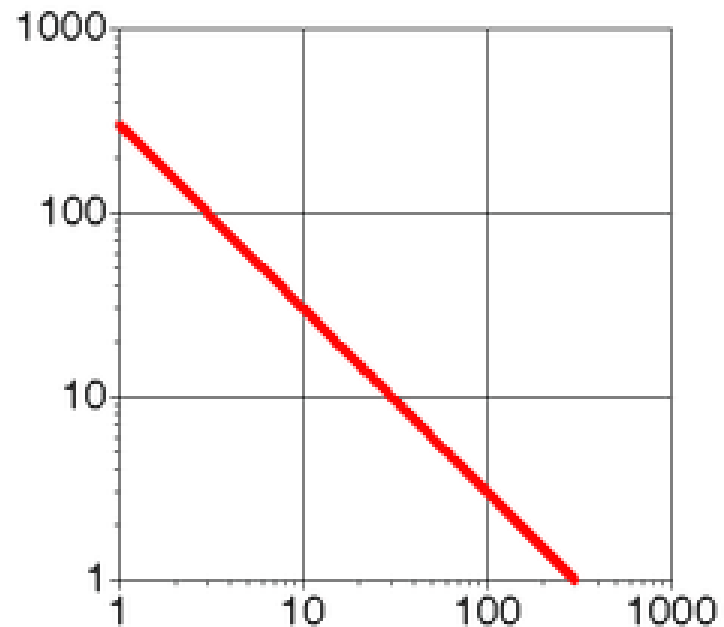
- From a discriminating point of view:
 - Queries use rather broadly defined, frequently occurring terms
 - It is the more specific terms that are particularly important in identifying relevant material

$$idf(t_k) = \log \frac{N}{D_k}$$

Linear and log scale



$$\frac{N}{D_k}$$



$$\log \frac{N}{D_k}$$

TF – IDF Weighting Schemes

$$w_{kd} = f_{kd} \left(\log \frac{N}{D_k} \right)$$

Problem if D_k is zero

TF – IDF Weighting Schemes

$$w_{kd} = f_{kd} \left(\log \frac{N}{D_k} \right)$$

Problem if D_k is zero

$$w_{kd} = f_{kd} \left(\log \frac{N+1}{D_k+1} \right) **$$

TF – IDF Weighting Schemes

$$w_{kd} = f_{kd} \left(\log \frac{N}{D_k} \right)$$

Problem if D_k is zero

$$w_{kd} = f_{kd} \left(\log \frac{N+1}{D_k+1} \right)^{**}$$

$$W_{kd} = \underbrace{(1 + \log(f_{kd}))}_{\text{TF}} \underbrace{\left(\log \frac{(N - D_k) + 0.5}{D_k + 0.5} \right)}_{\text{IDF}}$$

Moderates extreme values
(think: web spam!)

Better probabilistic interpretation:
(ratio between docs not containing the term and those with the term)

Best Match Retrieval Algorithm (with weights)

- Given what we know now about term weighting, we can update the Best-Match retrieval algorithm from before:

- Best-Match

for each document I, Score(I) = 0; I = 1 to N

for each query term t_k

Search the vocabulary list

Pull out the postings list

for each document J in the list,

Score(J) = Score(J) + w_{kd}